DC MOTOR SPEED STEP RESPONSE

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Apply a voltage step to the DC motor from a non-zero operating point and log the input, output and time data to the MATLAB workspace.	☐
2	Determine the steady-state gain and time constant of the motor from the measured step response, and relate them to the rise time and settling time.	☐
3	Assess how well a first-order transfer function represents the motor by repeating the test for different step sizes and directions.	☐

2.0 Hardware Set-Up

This practical exercise builds on the hardware and algorithm set-up described in the practical exercise, DC Motor Speed Measurement using an Encoder, which details the operation of a DC motor with an encoder and the measurement of motor speed in revolutions per minute (RPM). The set-up is not repeated here.

A discrete-time low-pass filter on the measured speed, as designed in the practical exercise, Discrete-Time Low-Pass Filter Design for DC Motor Speed Measurement, is recommended. A filter time constant of 0.1 s or less keeps the measurement noise low without noticeably slowing the measured response.

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. A step in voltage is applied to the motor in open loop, the response is logged to the MATLAB workspace, and the gain and time constant are read from the response. This links the measured behaviour of the motor to the first-order transfer function covered in the ACE-CORE learning guide, Response Analysis of First-Order Transfer Functions.

Background: The First-Order Step Response

When the inductance of the motor winding is small, the speed of a DC motor responds to the applied voltage as a first-order system, with the transfer function:

$$G(s) = \frac{\Omega(s)}{U(s)} = \frac{K}{\tau s + 1}$$

where *K* is the steady-state gain (RPM/V) and *τ* is the time constant (s). For a step change in voltage of size Δ*u*, applied when the motor is running steadily at speed ω_0, the speed responds as:

$$\omega(t) = \omega_0 + K\,\Delta u(1 - e^{-t/\tau})$$

The response has no overshoot. It reaches 63.2% of its final change after one time constant, and the standard time-domain measures follow directly from *τ*:

$$t_r \approx 2.2\,\tau \quad (10\text{–}90\%), \qquad t_s \approx 4\,\tau \quad (2\%)$$

Build the Step Test Model

The step is applied from a non-zero operating point rather than from rest. Below a certain voltage the motor does not turn at all because of static friction (the dead-zone), so a step from zero would mix this nonlinear behaviour with the dynamics being measured. In the Simulink model, a Step block defines the voltage applied to the motor, and a Gain of 255/5 converts it into the PWM value, see Figure 1.

[Screenshot to insert] *Simulink model: Step block (V) → Gain 255/5 → PWM output to the H-bridge; encoder speed (RPM) → low-pass filter → Scope; Clock → Scope. Scopes log u, y and t.*

Figure 1: *Simulink Model for the Open-Loop Step Test, Showing the Step Input in Volts (a), the PWM Output (b) and the Measured Speed (c)*

1. Open the speed measurement model from the practical exercise, DC Motor Speed Measurement using an Encoder, and save it under a new name.
2. Replace the Constant driving the PWM output with a Step block. Set the Step time to 1 s, the Initial value to 2 V and the Final value to 4 V.
3. Add a Gain of 255/5 between the Step block and the PWM output, so the step is defined in volts.
4. Log the step voltage, the measured speed and the time to the MATLAB workspace with the Variable names *u*, *y* and *t*, using the Save format Structure With Time, as in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares.
5. Set the Stop Time to 3 s and the sample time to 0.01 s, and run the model on the hardware. Confirm that *u*, *y* and *t* appear in the workspace.

Measure the Gain and Time Constant

Example results are shown in Figure 2. They are simulated with the motor model identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares (K = 28 RPM/V, τ = 0.15 s), with encoder quantisation and a 0.05 s low-pass filter added, so they show the shape to expect rather than measured data. The speed rises from 56 RPM to 112 RPM with no overshoot.

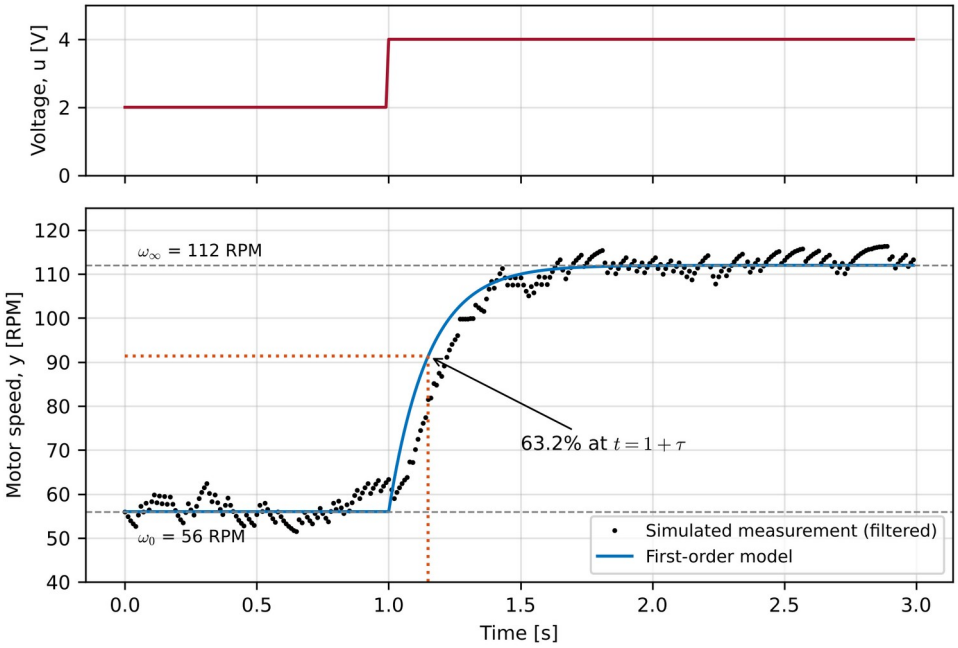


Figure 2: *Simulated Example of the Input Voltage (Top) and Motor Speed (Bottom) for a 2 V to 4 V Step, using the A1.4 Example Model*

The MATLAB script below calculates the gain, time constant, rise time and settling time from the logged data, and overlays the first-order model on the measured response. Run it after each step test.

MATLAB Script: measuring the gain, time constant, rise time and settling time

```
% Run after the step test, with u, y and t in the workspace

%% Load the logged data (Scope save format: Structure With Time)
u_k = u.signals.values(:);          % input voltage [V]
y_k = y.signals.values(:);          % measured motor speed [RPM]
t_k = t.signals.values(:);          % time [s]
t_step = 1;                         % time of the input step [s]

%% Initial and final values, averaged over 0.5 s windows
i0 = t_k > t_step-0.5 & t_k < t_step;
i1 = t_k > t_k(end)-0.5;
u0 = mean(u_k(i0));   u1 = mean(u_k(i1));
y0 = mean(y_k(i0));   y1 = mean(y_k(i1));
K  = (y1-y0)/(u1-u0)                % steady-state gain [RPM/V]

%% Normalised response after the step
idx = t_k >= t_step;
tn  = t_k(idx) - t_step;            % time since the step [s]
yn  = (y_k(idx)-y0)/(y1-y0);        % 0 before the step, 1 at steady state
tau = tn(find(yn >= 0.632,1))       % time constant [s]
tr  = tn(find(yn>=0.9,1)) - tn(find(yn>=0.1,1))   % rise time [s]
ts  = tn(find(abs(yn-1)>0.02,1,'last')+1)        % settling time [s]

%% Compare the measured response with the first-order model
G = tf(K,[tau 1]);
y_model = y0 + (u1-u0)*step(G,tn);
plot(t_k,y_k,'k.', tn+t_step,y_model,'b'); grid on;
xlabel('Time [s]'); ylabel('Motor speed [RPM]');
legend('Measured speed','First-order model','Location','southeast');
```

Measurement noise can make the speed cross the 63.2% or 2% thresholds early. If the values look inconsistent, check that the low-pass filter is in place, and compare them with the ratios in the equations above: the rise time should be close to 2.2τ and the settling time close to 4τ.

Table 2: *Expected Results for the A1.4 Example Model (Simulated)*

Quantity	Symbol	Expected value	Basis
Steady-state gain	K	28 RPM/V	(112 − 56) / (4 − 2)
Time constant	τ	0.15 s	63.2% of the change
Rise time (10–90%)	t_r	0.33 s	$\approx 2.2\tau$
Settling time (2%)	t_s	0.59 s	$\approx 4\tau$

Test the Linearity of the Model

A linear model predicts the same K and τ for every step size and direction. Real motors depart from this because of friction, the dead-zone and changes in supply voltage under load. Repeat the step test for the cases in Table 3, changing only the Initial value and Final value of the Step block.

1. Run the small step (2 V to 3 V) and the large step (1.5 V to 4.5 V), and record the results in Table 3.

2. Run the downward step (4 V to 2 V). For a downward step, y_1 is smaller than y_0, but the script still normalises the response from 0 to 1.
3. Finally run a step from rest (0 V to 4 V) and compare it with the other cases.

Table 3: *Step Test Results for Your Motor*

Step	Δu (V)	K (RPM/V)	τ (s)	t_r (s)	t_s (s)
2 V → 4 V	2				
2 V → 3 V	1				
1.5 V → 4.5 V	3				
4 V → 2 V	−2				
0 V → 4 V	4				

Compare your results with each other and with the expected values in Table 2. The steps that start above the dead-zone should give similar values of K and τ. The step from rest usually gives a lower apparent gain and a delayed start, because part of the voltage is used to overcome static friction before the motor moves. Explain any other differences you observe, and decide which operating range a single first-order model can represent well.

4.0 Concluding Remarks

This practical exercise measured the open-loop step response of a DC motor and related it to a first-order transfer function. Using MATLAB, Simulink and ACE-Lab hardware, the steady-state gain was found from the change in speed divided by the change in voltage, and the time constant from the time taken to reach 63.2% of the change. The rise time and settling time were then shown to follow from the time constant.

Repeating the test for different step sizes and directions showed the limits of a linear model. Steps taken above the dead-zone give consistent parameters, which is why the motor model used to design a controller should be identified around the speed at which the controller will operate.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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